

ADEN GUO

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TECH STACK

- **Languages & Data:** Python, TypeScript, JavaScript, Rust, C++, SQL, NumPy, Pandas, scikit-learn, PyTorch, OpenCV
- **Web & Frameworks:** React, Next.js (App Router), SvelteKit, Node.js, FastAPI, Tailwind CSS, Tauri, Drizzle ORM, Bun
- **Databases & Retrieval:** PostgreSQL, pgvector (hybrid semantic search), SQLite, Redis, Supabase, Neon
- **Cloud & Infrastructure:** Azure, AWS S3, Vercel, Railway, Docker, CI/CD pipelines, Distributed & event-driven systems
- **AI & LLM Engineering:** Anthropic / Claude API, OpenAI, Vercel AI SDK, OpenRouter, RAG & vector retrieval, Prompt caching, MCP & agentic CLI tooling
- **Systems & DevOps:** Linux, Bash, Git & GitHub, Docker, VS Code
- **GIS & Survey:** ArcGIS, QGIS, PostGIS, GeoPandas, GDAL/PROJ, RTK GNSS workflows, Agisoft Metashape

EDUCATION

Bachelor of Science: Geomatics Engineering [Graduated]

University of Calgary - Calgary, AB

Relevant Skills:

- Photogrammetry, 3D modeling, and digital imaging workflows
- Geospatial data analysis, error modeling, and quality control
- Spatial database design, SQL, and geospatial data mining
- Project management, technical reporting, and collaborative problem-solving
- Applied geomatics workflow through capstone and ENGO 500 field survey camp

PROFESSIONAL EXPERIENCE

Engineering Capstone Project Lead - *University of Calgary*

Sep 2024 - Apr 2025

- Compared RTK GNSS, range finder, and odometer techniques to assess mapping accuracy and efficiency.
- Operated drones for photogrammetric capture and executed spatial intersection workflows for 3D modeling.
- Processed and validated geospatial datasets with photogrammetry tools and Python-based analysis pipelines.
- Authored and presented the final report at the SaGES Project Presentation.

Shift Lead / Team Trainer - *Gongcha*

Nov 2020 - May 2024

- Trained and mentored staff to improve consistency and teamwork.
- Balanced service demands and operational tasks under time constraints.
- Strengthened team coordination, problem-solving, and day-to-day execution.

PROJECTS

Glass Atlas — Source-Grounded RAG Knowledge Site

- Deployed SvelteKit 2 (Svelte 5 runes) editorial site that turns a personal note library into a conversational, citation-backed assistant.
- Engineered hybrid semantic + lexical retrieval over Neon PostgreSQL with pgvector (cosine similarity) and Drizzle ORM, with confidence-tiered fallbacks that keep every answer grounded in source notes.
- Built streaming RAG chat through OpenRouter (Gemini 2.0 Flash) on text-embedding-3-small 1536-dim vectors, with strict-TypeScript end-to-end typing.
- Shipped on Railway with a Bun HTTP server, Auth.js GitHub OAuth, and private S3-compatible storage via presigned URLs.

Weather & Wellness — UBC Research Platform

- Production multi-tenant research platform for a UBC Psychology lab where research assistants run validated psychological instruments with anonymous participants.
- Built a Next.js 15 / React 19 frontend and FastAPI backend that auto-scores validated instruments (digit span, loneliness, depression, anxiety) in real time.
- Implemented weather–psychology mixed-effects models with statsmodels, NumPy, and Pandas over per-lab isolated datasets.
- Architected per-lab data isolation on Supabase Postgres/Auth with Upstash Redis, shipped to production at ubcpsych.com.